DNN Programming Assignment 2

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Chosen settings h = 48; d = 96; model seeds 7, 21, 42.

Assumptions and choices. As clarified by the professor, I chose h, d and the three seeds instead of using the variants table. My ID ends in 4, so I used the clipping ablation. Since no attention example was supplied, I used the small Q/K/V example on page 4. I chose 8 epochs for HAR, 12 for adding/clipping and 600 steps for each language-model run. Models within a comparison get the same budget.

Problem 1: Data, cell verification and HAR

For UCI HAR, I used the nine inertial channels and held out four training subjects for validation for each seed [1]. Mean and standard deviation were fitted only on the remaining training data. Adam uses lr 0.001 and batch 128. Adding has 10,000/1,000/1,000 examples with data seeds 0/1/2. I kept the epoch with the lowest validation loss; the test set was not used to choose it.

Table 1. Cell verification and counts at input 9, h = 48. Count = g*h*(9+h+1), or g*h*(9+h+2) for two biases. PyTorch GRU reset-after biases are retained for exact parity [3].

Cell	One bias	Two biases	Max sequence error
RNN	2784	2832	1.19e-07
LSTM	11136	11328	8.94e-08
GRU	8352	8496	2.38e-07

Table 2. HAR test mean +/- sample SD over three seeds; parameter counts include the 294-parameter classifier head. Seconds cover training loops only.

Model	Accuracy	Macro F1	Params	s/epoch
GRU	0.877 +/- 0.019	0.877 +/- 0.019	8790	0.881
LSTM	0.858 +/- 0.011	0.856 +/- 0.013	11622	0.972
RNN	0.655 +/- 0.067	0.623 +/- 0.088	3126	0.301

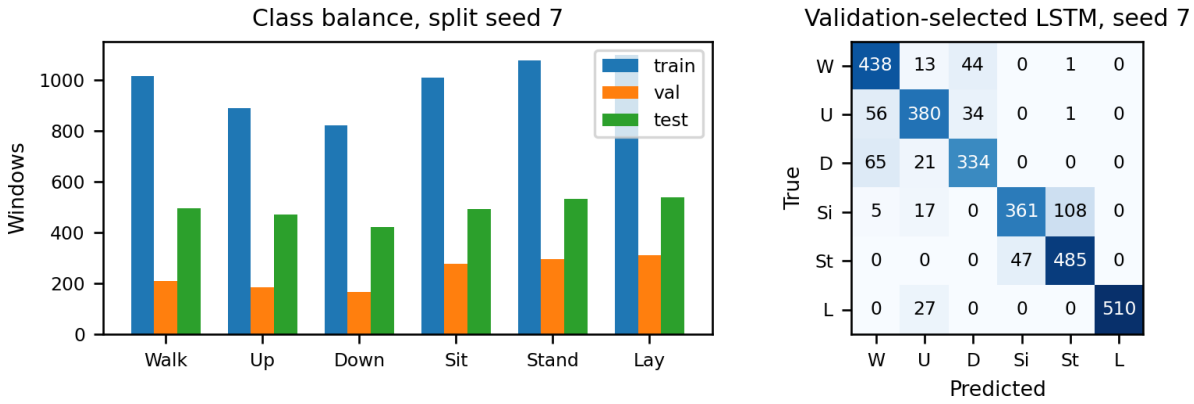


Figure 1. Class balance and validation-selected LSTM confusion matrix. W/U/D/Si/St/L denote walk/upstairs/downstairs/sit/stand/lay. Selection uses mean validation loss by family, then the lowest validation-loss seed; GRU instead has the highest retrospective test mean.

Problem 1: Adding and state gradients

Table 3. Test MSE mean +/- sample SD; the second line gives solved runs out of three, using strict MSE < 0.05. Finite-test constant-1 MSE is shown in Figure 2; its theoretical expectation is 1/6.

Model	T = 50	T = 100	T = 200	T = 400
RNN	0.1749 +/- 0.0007; 0/3	0.1699 +/- 0.0001; 0/3	0.1683 +/- 0.0003; 0/3	0.1541 +/- 0.0007; 0/3
LSTM	0.1735 +/- 0.0005; 0/3	0.1697 +/- 0.0002; 0/3	0.1687 +/- 0.0001; 0/3	0.1538 +/- 0.0005; 0/3
GRU	0.0098 +/- 0.0008; 3/3	0.0408 +/- 0.0040; 3/3	0.1665 +/- 0.0021; 0/3	0.1533 +/- 0.0009; 0/3

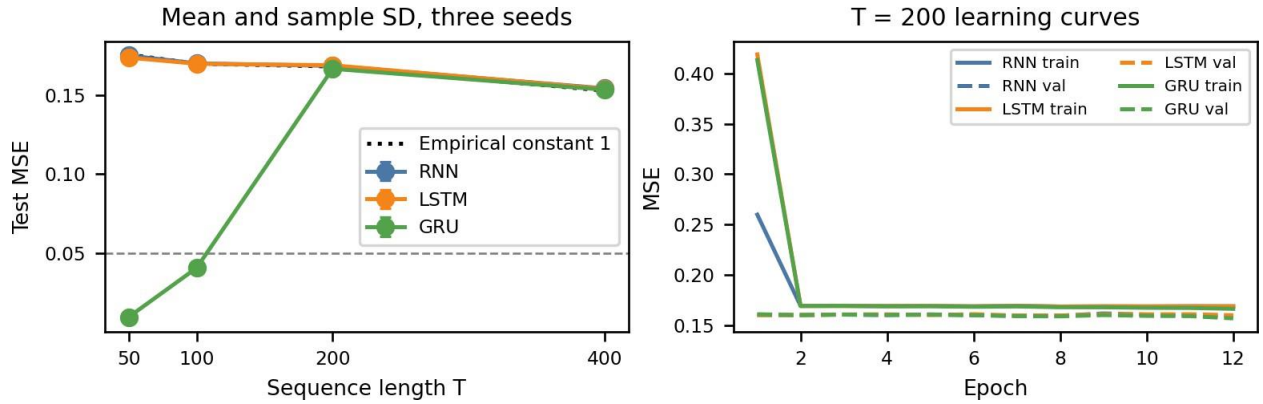


Figure 2. Test MSE against length and T = 200 training/validation curves. All models receive equal budgets. Test baseline variation across T comes from finite generated samples; raw MSE decreases alone do not establish better memory.

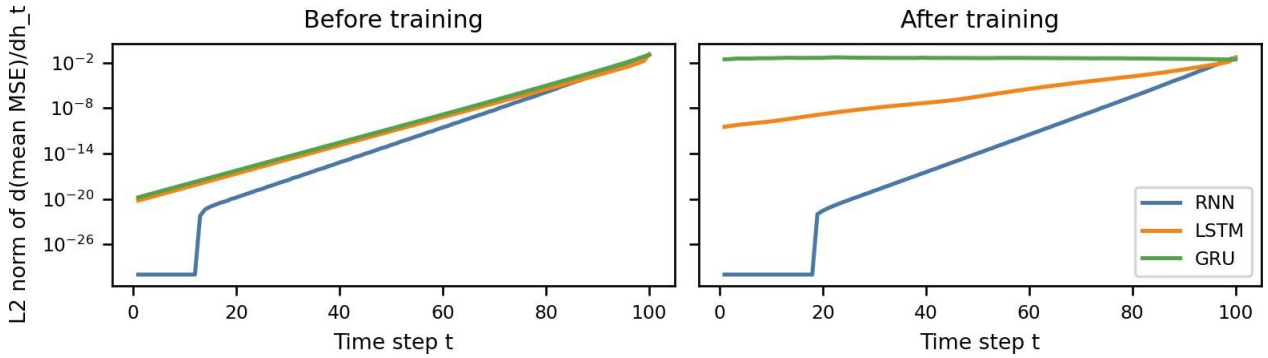


Figure 3. Mean state-gradient L2 norms over model seeds, measured on the same fixed T = 100 diagnostic examples before and after training. Framework cells are unrolled with copied weights, retaining actual states used by the recurrence. Display floor 1e-30; raw zeros remain in CSV. Clipping is not applied to this diagnostic.

Problem 1: Analysis and individual ablation

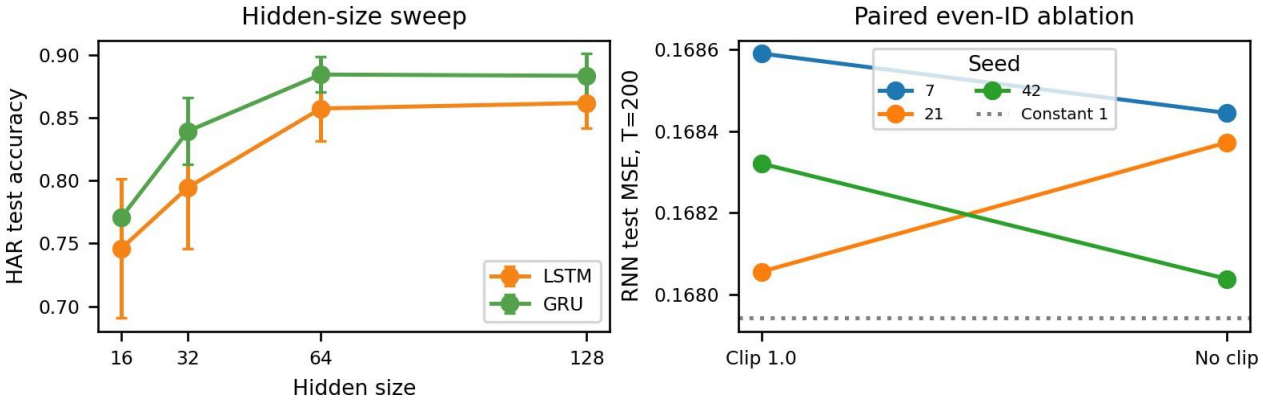


Figure 4. LSTM/GRU hidden-size sweep and paired even-ID RNN clipping experiment at T = 200. Initialisation, minibatch order and 12-epoch budget are paired for clipping versus none.

The small errors in Table 1 show that the three cells match the library versions. Figure 1 shows the class counts and the confusion matrix for the model selected using validation loss.

Figure 3 helps explain the adding results. After training at T = 100, the first/last hidden-gradient norms are 0.0277/0.0260 for GRU, 3.23e-11/0.0488 for LSTM, and 0.00e+00/0.0426 for RNN. GRU keeps a stronger gradient at early steps. The early RNN values underflow to zero in float32. LSTM hidden-state gradients also become very small, but this plot does not measure its separate cell-state memory path.

In Table 3 and Figure 2, GRU solves 3/3 runs at T = 50 and 3/3 at T = 100, but 0/3 at T = 200 and 0/3 at T = 400. RNN and LSTM stay near the constant baseline after 12 epochs. I would not conclude that LSTM cannot solve this task: it did not learn the solution with these settings. More training or a separate initialisation experiment would be needed to check that.

Clipping gives mean MSE 0.1683, compared with 0.1683 without clipping. There is little difference here. Clipping limits large parameter gradients, but does not solve the loss of gradients across time. Figure 4 also shows the hidden-size sweep; a larger hidden state is useful only if its accuracy gain justifies its extra parameters and training time.

For a wearable prototype, I would start with GRU based on Table 2. It gives 87.67% mean accuracy with 8,790 parameters, compared with 85.77% and 11,622 for LSTM, and trains faster in these runs. RNN is smaller at 3,126 parameters, but its accuracy is only 65.55%. This recommendation uses the observed test results; the validation-based choice remains LSTM. Actual inference time, RAM and energy still need checking. HAR seeds change both training and validation subjects, so the reported SD includes both sources of variation; test subjects stay fixed.

Problem 2: Transformer and HAR

Problem 2 begins with attention and the encoder block (Tasks 1-2), followed by the HAR classifier (Task A). The character model is on page 5 and its ablations are on page 6.

Manual attention passes row-normalisation, future-zero and functional-reference checks; maximum absolute error 5.55e-16. Copied-weight multi-head attention matches its library reference within 2.50e-16. Tests also verify shapes, sinusoidal positions, no-position patch permutation invariance and end-to-end causal isolation. Library attention appears only in verification.

At d = 96 and d_ff = 384, block matrix weights = 4d^2 + 2*d*d_ff = 110,592. Projection/FFN biases add 864, and two affine LayerNorms add 384: total 111,840 per block. Multi-head attention alone has 37,248 parameters. Blocks use pre-LayerNorm, GELU, two residual paths and dropout 0.1 [4].

Paper Calculations Exam. Choose head dimension 2 and Q = K = V = [[1,0],[0,1],[1,1]]. Let s = 1/sqrt(2), a = exp(s) = 2.028114982. Scaled score rows are [s,0,s], [0,s,s], [s,s,2s]. Softmax rows are [a,1,a]/(2a+1), [1,a,a]/(2a+1), [1,1,a]/(a+2). Multiplication by V gives Table 4.

Table 4. Hand and manual-code attention outputs agree at 1e-12 tolerance in float64; framework output also agrees. Full Q/K/V, scores and weights appear in the completed notebook.

Query	Calulcations on Paper	Manual-code output
1	[0.80222419, 0.59888791]	[0.80222419, 0.59888791]
2	[0.59888791, 0.80222419]	[0.59888791, 0.80222419]
3	[0.75174492, 0.75174492]	[0.75174492, 0.75174492]

Task A - HAR encoder classifier. Table 5 reports test mean +/- sample SD using identical subject splits/statistics, Adam, batch and 8-epoch budget. Chosen architecture: sixteen non-overlapping 8-step patches plus CLS and two encoder blocks. The recurrent family is selected by validation.

Model / position	Accuracy	Macro F1	Params	s/epoch
learned	0.895 +/- 0.016	0.894 +/- 0.017	233190	1.448
none	0.923 +/- 0.007	0.923 +/- 0.008	231558	1.426
sinusoidal	0.912 +/- 0.001	0.912 +/- 0.001	231558	1.428
LSTM recurrent	0.858 +/- 0.011	0.856 +/- 0.013	11622	0.972

In Table 5, sinusoidal positions change mean accuracy by -1.108 percentage points compared with no positions. Three seeds are too few to claim a reliable general difference. The no-position model still sees the order within each 8-reading patch, so this result does not mean that temporal order is unimportant.

Problem 2 - Task B: Language model

Table 6. Main causal character model: vocabulary 65 from the whole Tiny Shakespeare text [2]; chronological 90/10 split; seed 7, context 128, d = 96, four heads, four blocks, Adam 0.001, batch 32, 600 steps. All 460,097 weights start randomly. Results use final-step weights; full validation counts each target once in reset-context chunks.

Predictor	Validation nats	Bits per character
Uniform 65-way	4.1744	6.0224
Trained causal model	2.0238	2.9198

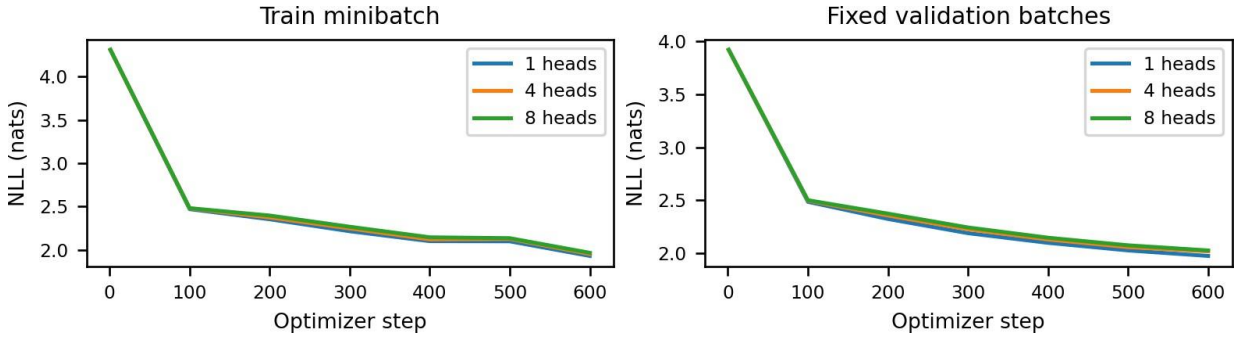


Figure 5. Equal-budget head-count learning curves. Curve validation uses fixed sampled batches; Table 6 uses full held-out text. The 600-step run does not reach the brief sanity range; the learning curve is still decreasing, so training budget limits the result.

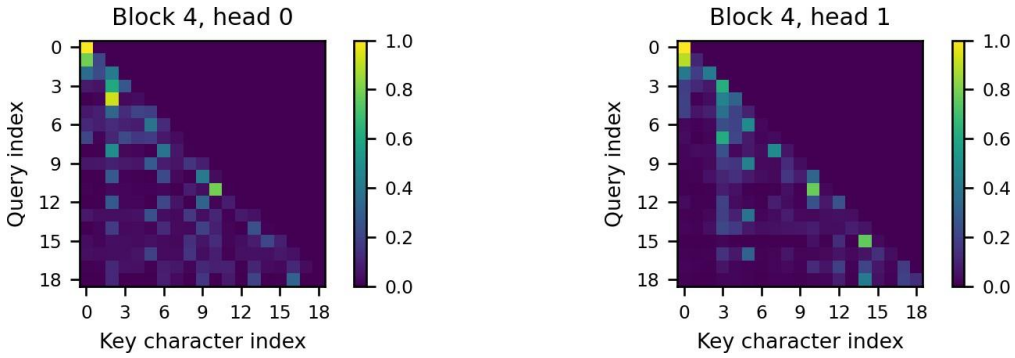


Figure 6. Last-block heads on “To be, or not to be”. Excluding the first query, mean attention-weighted backward distance is 4.06 characters for head 0 and 3.38 for head 1: head 0 places more weight on distant context in this sentence. Future weights are exactly zero. One sentence does not establish syntactic roles.

The loss is still falling in Figure 5. Table 6 gives the loss over the full validation text, while Figure 6 shows that the two heads use earlier characters differently.

Generated samples use top-k 50 and exactly 500 new characters at each temperature. The 0.2/0.8/1.5 samples contain 31/39/57 distinct characters. Low temperature favours repeated high-probability fragments; high temperature introduces more varied spellings and punctuation. Full samples are in generated_samples.txt and the notebook.

Problem 2: Analysis and ablations

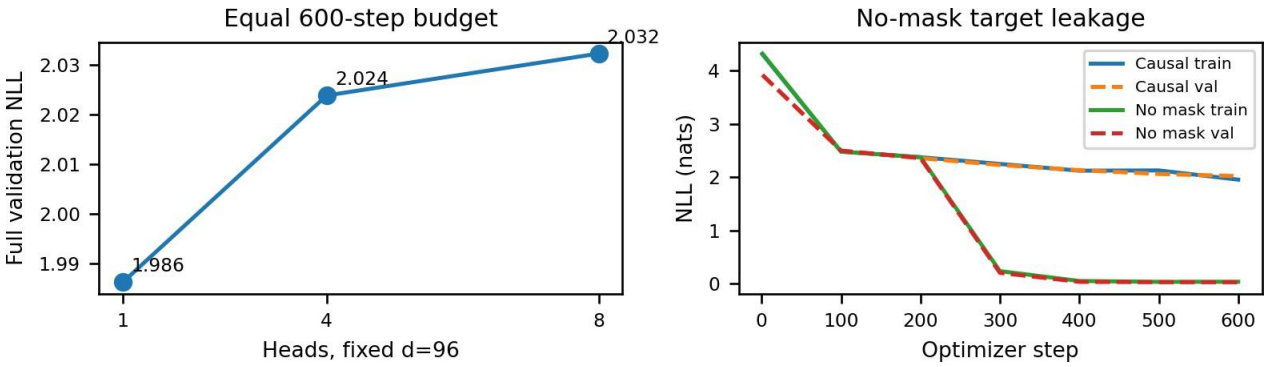


Figure 7. Head and mask ablations use seed 7 only, matched training windows and 600 steps each. Heads 1/4/8 give NLL 1.9863/2.0238/2.0323. Parameters remain 460,097. These are single-seed results, not three-seed estimates; they cannot establish a general ranking.

Figure 7 shows that without a causal mask, teacher-forced validation falls to 0.0232 nats because future input exposes the target. Restoring the mask gives 8.5082 nats, worse than uniform. The saved no-mask sample collapses to 499 newlines and one question mark; its low leaked loss is not a valid causal quality improvement.

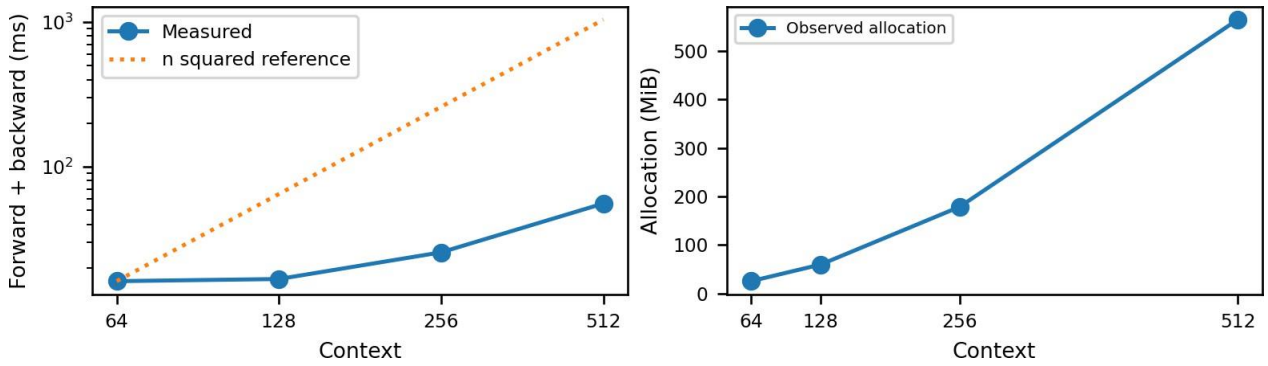


Figure 8. Fixed batch 8, width, heads and layers. Each context uses 3 warmups and 10 forward/backward steps, excluding optimiser updates. From context 64 to 512, time grows 3.4x and observed allocated memory grows 22.2x. Memory is sampled at synchronized forward/backward checkpoints and may miss a transient peak.

Figure 8 does not show a pure quadratic increase in step time. Attention has an n-squared term, but the projections, feed-forward layers and fixed overheads also take time. Parallel execution affects the measured trend as well. Memory growth is closer to quadratic. The dotted curve shows the theoretical reference, not another measured run.

The full grid contains 75 recurrent runs, 9 Transformer HAR runs, 4 language conditions and 4 context measurements. Summed measured training-loop time is 19.6 minutes, excluding validation/tests/plots and interrupted partial work. Timing varies between runs, so small differences need cautious interpretation. No bonus experiment is claimed.

AI use and references

AI use statement: I used OpenAI Codex for concept explanations and debugging. Assistance in report preparation. The numerical results come from the saved program runs. No pretrained weights or additional datasets were used.

[1] Reyes-Ortiz et al. Human Activity Recognition Using Smartphones. UCI Machine Learning Repository, 2013. DOI 10.24432/C54S4K. Official dataset.

[2] Karpathy. char-rnn repository, Tiny Shakespeare input text. Source text.

[3] PyTorch. Recurrent and functional attention documentation. GRU convention; attention verification reference.

[4] Vaswani et al. Attention Is All You Need. 2017. arXiv:1706.03762. Paper.